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Project

Custom System Monitor

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Class: 4

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Contents

- 1 Introduction**
- 2 Environment Setup**
- 3 Creating the Script File**
- 4 Code Explanation**
 - 4.1 CPU Usage Monitoring Function**
 - 4.2 Memory Usage Monitoring Function**
 - 4.3 Disk Usage Monitoring Function**
 - 4.4 Network Activity Monitoring Function**
 - 4.5 Battery Status Monitoring Function**
 - 4.6 Input/Output Status Monitoring Function**
- 5 Conclusion**
- 6 References**

1. Introduction

In this report, we will explore how to monitor the performance of Linux operating systems using Bash scripting. We will create a set of functions that allow us to gather valuable information about resource usage on the system, including CPU, memory, disk space, network activity, battery status, and input/output status. These tools aim to help us identify potential issues and improve overall system management and performance.

2. Environment Setup

To start using these tools, we will first install the **sysstat** package, which provides many useful commands for monitoring system performance, including battery status. we can install the package using:

```
abrar@abrar-VirtualBox: ~/Desktop
abrar@abrar-VirtualBox:~/Desktop$ sudo apt install sysstat
[sudo] password for abrar:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package sysstat
```

Furthermore, we will also install the **iftop** package to monitor network activity:

```
abrar@abrar-VirtualBox:~/Desktop$ sudo apt install ifstat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ifstat
0 upgraded, 1 newly installed, 0 to remove and 82 not upgraded.
Need to get 22.4 kB of archives.
After this operation, 62.5 kB of additional disk space will be used.
Get:1 http://sa.archive.ubuntu.com/ubuntu jammy/universe amd64 ifstat amd64 1.1-8.2 [22.4 kB]
Fetched 22.4 kB in 1s (27.2 kB/s)
Selecting previously unselected package ifstat.
(Reading database ... 203864 files and directories currently installed.)
Preparing to unpack .../ifstat_1.1-8.2_amd64.deb ...
Unpacking ifstat (1.1-8.2) ...
Setting up ifstat (1.1-8.2) ...
Processing triggers for man-db (2.10.2-1) ...
abrar@abrar-VirtualBox:~/Desktop$
```

3. Creating the Script File

After preparing all the necessary functions, we will now create the monitoring.sh script file using the nano text editor:

```
abrar@abrar-VirtualBox: ~/Desktop
abrar@abrar-VirtualBox:~/Desktop$ nano monitoring.sh
```

The Script:-

```
abrar@abrar-VirtualBox: ~/Desktop
GNU nano 6.2 monitoring.sh
echo " CPU Usage"
mpstat 1 1

echo ""
echo "Memory Usage"
free -h

echo ""
echo " Disk Usage"
df -h

echo ""
echo " I/O Stats "
iostat -xz 1

echo ""
echo " Network Activity"
timeout 15 ifstat

echo ""
echo "Battery and Power Status"
acpi -i

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute   ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify   ^_ Go To Line
```

After creating the monitoring.sh script file and adding the necessary functions, we need to make the script executable:

```
abrar@abrar-VirtualBox: ~/Desktop$ nano monitoring.sh
abrar@abrar-VirtualBox: ~/Desktop$ chmod +x monitoring.sh
abrar@abrar-VirtualBox: ~/Desktop$ ./monitoring.sh
```

4. The Code

4.1 CPU Usage Monitoring Function

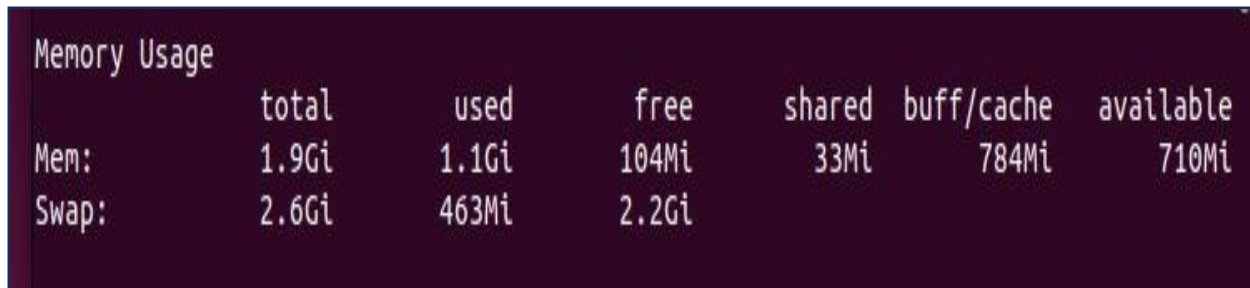
This function uses the mpstat command to display detailed CPU usage.

```
abrar@abrar-VirtualBox: ~/Desktop$ ./monitoring.sh
CPU Usage
Linux 6.8.0-45-generic (abrar-VirtualBox)      الأربعاء 16 أكتوبر 2024
x86_64_ (6 CPU)

20:33:38      CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest
      %gnice   %idle
20:33:39    all    0.88    0.00    0.53    0.00    0.00    1.41    0.00    0.00
      0.00   97.18
Average:    all    0.88    0.00    0.53    0.00    0.00    1.41    0.00    0.00
      0.00   97.18
```

4.2 Memory Usage Monitoring Function

This function utilizes the free command to show information about memory usage in the system.

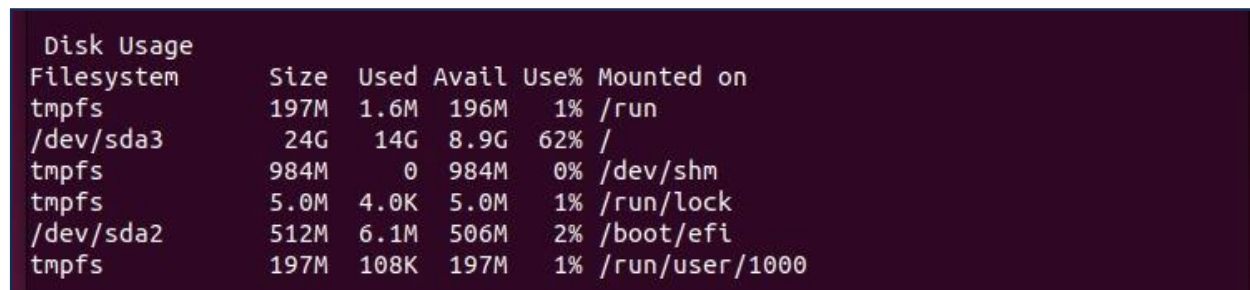


```
Memory Usage
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	1.1Gi	104Mi	33Mi	784Mi	710Mi
Swap:	2.6Gi	463Mi	2.2Gi			

4.3 Disk Usage Monitoring Function

This function employs the df command to present details about disk space usage.

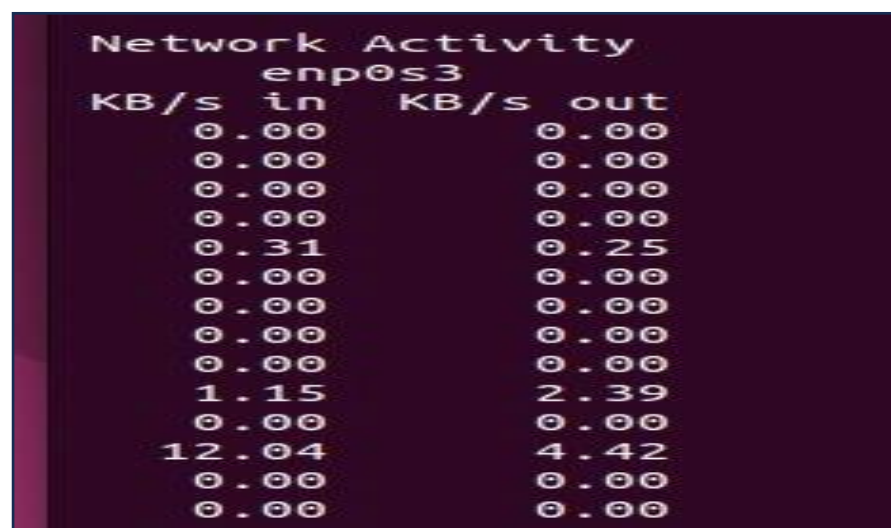


```
Disk Usage
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	197M	1.6M	196M	1%	/run
/dev/sda3	24G	14G	8.9G	62%	/
tmpfs	984M	0	984M	0%	/dev/shm
tmpfs	5.0M	4.0K	5.0M	1%	/run/lock
/dev/sda2	512M	6.1M	506M	2%	/boot/efi
tmpfs	197M	108K	197M	1%	/run/user/1000

4.4 Network Activity Monitoring Function

This function uses the ifstat command to display statistics about network activity.



```
Network Activity
```

enp0s3	
KB/s in	KB/s out
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.31	0.25
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
1.15	2.39
0.00	0.00
12.04	4.42
0.00	0.00
0.00	0.00

4.5 Battery Status Monitoring Function

This function reads battery charge levels and status.

```
Battery and Power Status
Battery 0: Charging, 78%, charging at zero rate - will never fully charge.
Battery 0: design capacity 5000 mAh, last full capacity 5000 mAh = 100%
abrar@abrar-VirtualBox: ~/Desktop$
```

4.6 Input/Output Status Monitoring Function

This function will monitor the input and output operations on the system. It can provide insights into how efficiently the system is handling data transfers.

```
I/O Stats
Linux 6.8.0-45-generic (abrar-VirtualBox)      الأربعاء 16 أكتوبر 2024
x86_64_ (6 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           2.52    0.07   9.26   0.74    0.00   87.42

Device            /s    wrqm/s    %wrqm  w_await  wareq-sz   r/s    rkB/s    rrqm/s    %rrqm  r_await  rareq-sz   w/s    wkB
req-sz    f/s  f_await   aqu-sz   %util    d/s    dKB/s    drqm/s    %drqm  d_await   da
loop0      0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop1      0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop10     0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop11     0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop12     0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop2      0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop3      0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
loop4      0.00  0.00    0.00    0.00    0.00    0.00  0.00    0.00    0.00    0.00    0.00    0.00  0.00
```

5. Conclusion

Through this report, we learned how to create a set of Bash functions for monitoring the performance of Linux operating systems. These tools will enable us to gather valuable information about resource usage, assisting in identifying issues and improving system management and performance. By using these tools, we can create automated scripts for continuous system performance monitoring.

6. References

- Use in problem solving <https://www.javatpoint.com/linux-commands>